Proof:

$$E(XY) = \sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j p_{ij}$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j p_i p_j$$

$$= \left(\sum_{i=1}^{n} x_i p_i\right)\left(\sum_{j=1}^{m} y_i p_i\right)$$

$$= E(X)\,E(Y)$$

If X and Y are independent

$$P_{ij} = p_i p_j$$

Continuous random variable:

Statement: If X and Y are independent random variables Then $E(XY) = E(X)\,E(Y)$

Proof:

Now $$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x, y)\,dx\,dy$$

$$= = \int_{-\infty}^{\infty} xf(x)\,dx \int_{-\infty}^{\infty} yf(y)\,dy$$

$$= E(X)\,E(Y)$$

Example 9.17

Two coins are tossed one by one. First throw is considered as X and second throw is considered as Y following joint probability distribution is given by,

X / Y	1	0	Total
1	0.25	0.25	0.5
0	0.25	0.25	0.5
Total	0.5	0.5	1

[getting Head is taken as 1 and Tail is taken as 0]

Verify $E(XY) = E(X)\,E(Y)$

Solution:

A random variable XY can take the values 0 and 1